



SunTech

RJ45 to RS 485 Adaptor Instructions



RJ45 to RS-485 ADAPTER INSTRUCTIONS

This adapter is required to connect the MT50 Remote to the larger capacity Solar Charge Controllers such as the ET6415BND. The adapter cable can be created by anyone with basic electrical wiring skills.

1. The ET6415BND ships with a USB to RS-485 adapter cable pictured below. Locate this cable and remove it from the package.

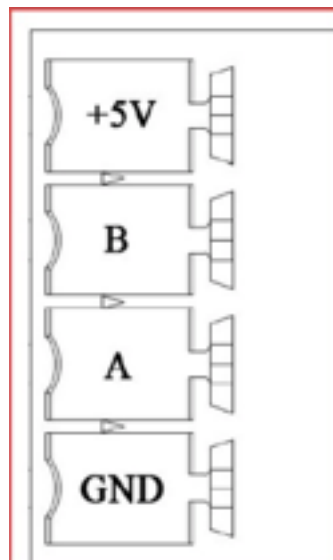


2. Cut off the RS-485 plug indicated above, leaving about 3 inches of wire attached.
3. Locate the RJ45 cable that was shipped with your MT50 remote or if you are using a longer RJ45 cable you will need that cable instead.

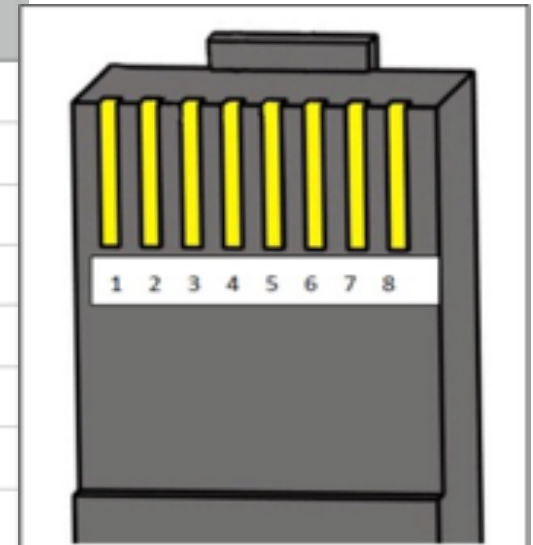


4. Cut off one of the RJ45 plugs and strip the cable exposing the wires. Connect the wires from the RS-485 plug you obtained in step 1 to the bare wires on the end of the RJ45 cable according to the following wire connection diagrams

Interface Pin Definition for RS485 on controller

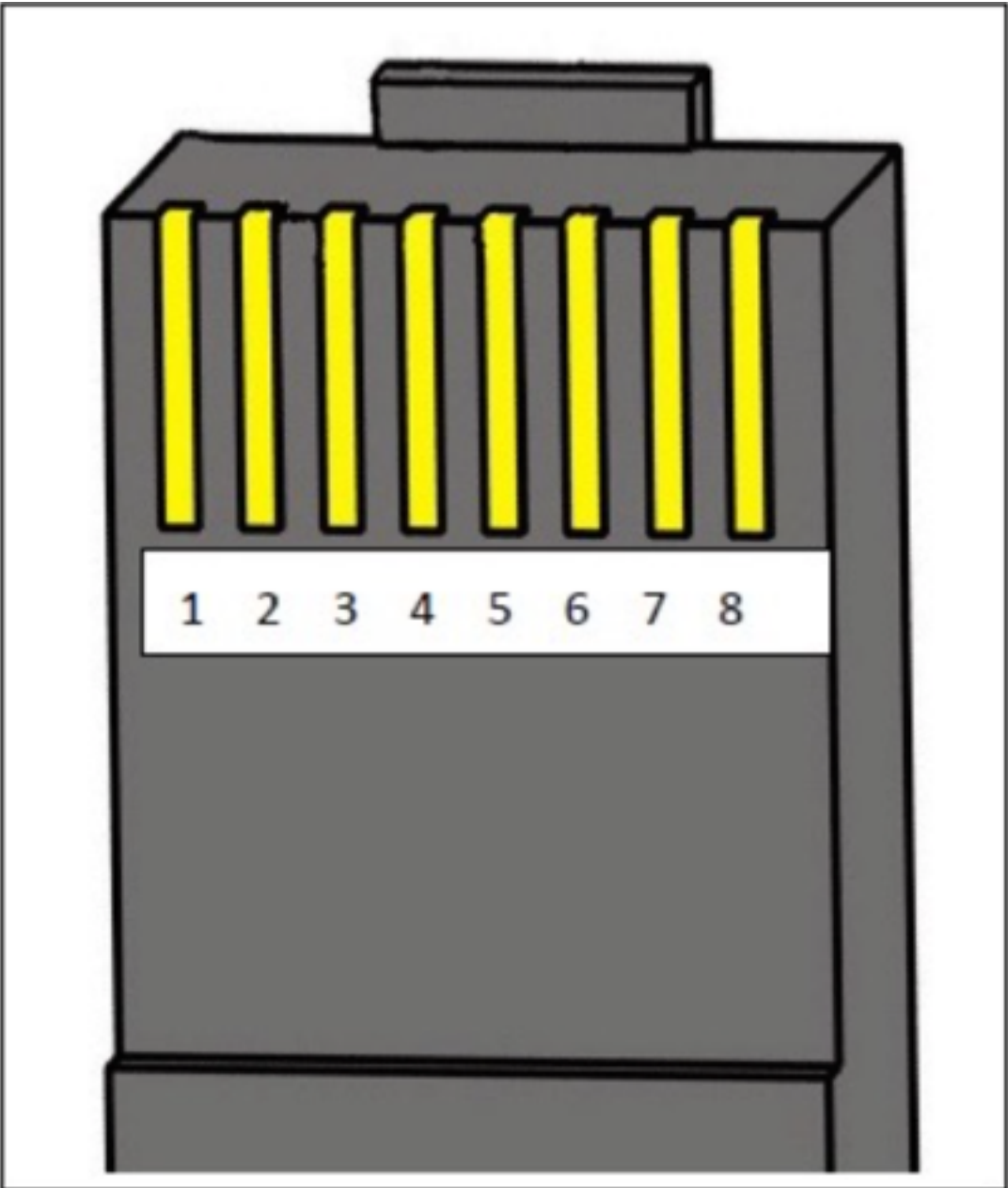


Pins	Define	Colour
1	Power supply output +5V	Orange
2	Power supply output +5V	White
3	RS-485-B	Black
4	RS-485-B	Red
5	RS-485-A	Green
6	RS-485-A	Yellow
7	Ground	Blue
8	Ground	Brown



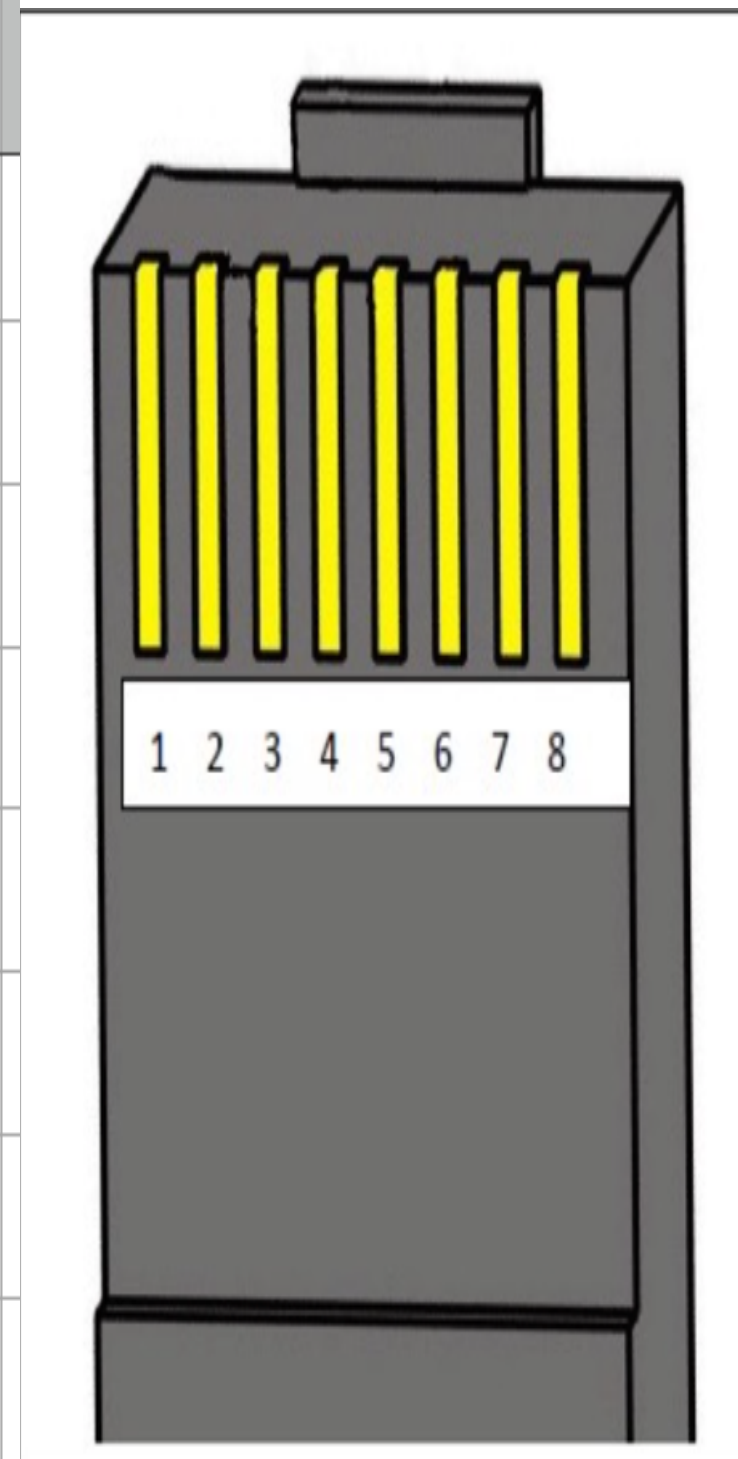
RS-485 Communication Cable Breakdown

Pins	Define	Colour
1	Power supply output +5V	Orange
2	Power supply output +5V	White
3	RS-485-B	Black
4	RS-485-B	Red
5	RS-485-A	Green
6	RS-485-A	Yellow
7	Ground	Blue
8	Ground	Brown



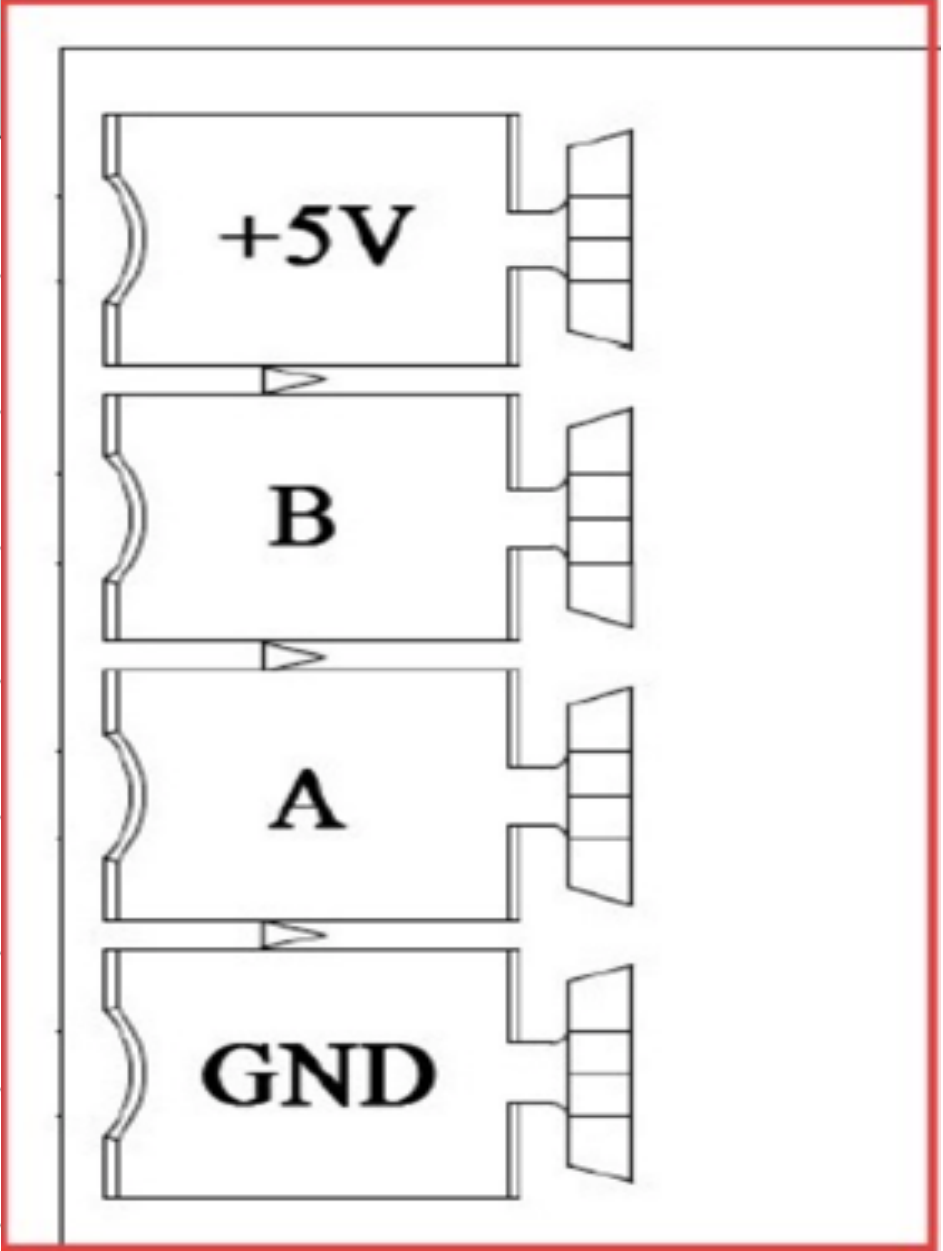
The diagram shows a grey 8-pin connector with a small tab on top. Eight yellow pins are visible, numbered 1 through 8 from left to right. The connector is shown in a perspective view.

Pins	Define	Colour
1	Power supply output +5V	Orange
2	Power supply output +5V	White
3	RS-485-B	Black
4	RS-485-B	Red
5	RS-485-A	Green
6	RS-485-A	Yellow
7	Ground	Blue
8	Ground	Brown



RJ-45 Communication Wire to RS-485 Plug

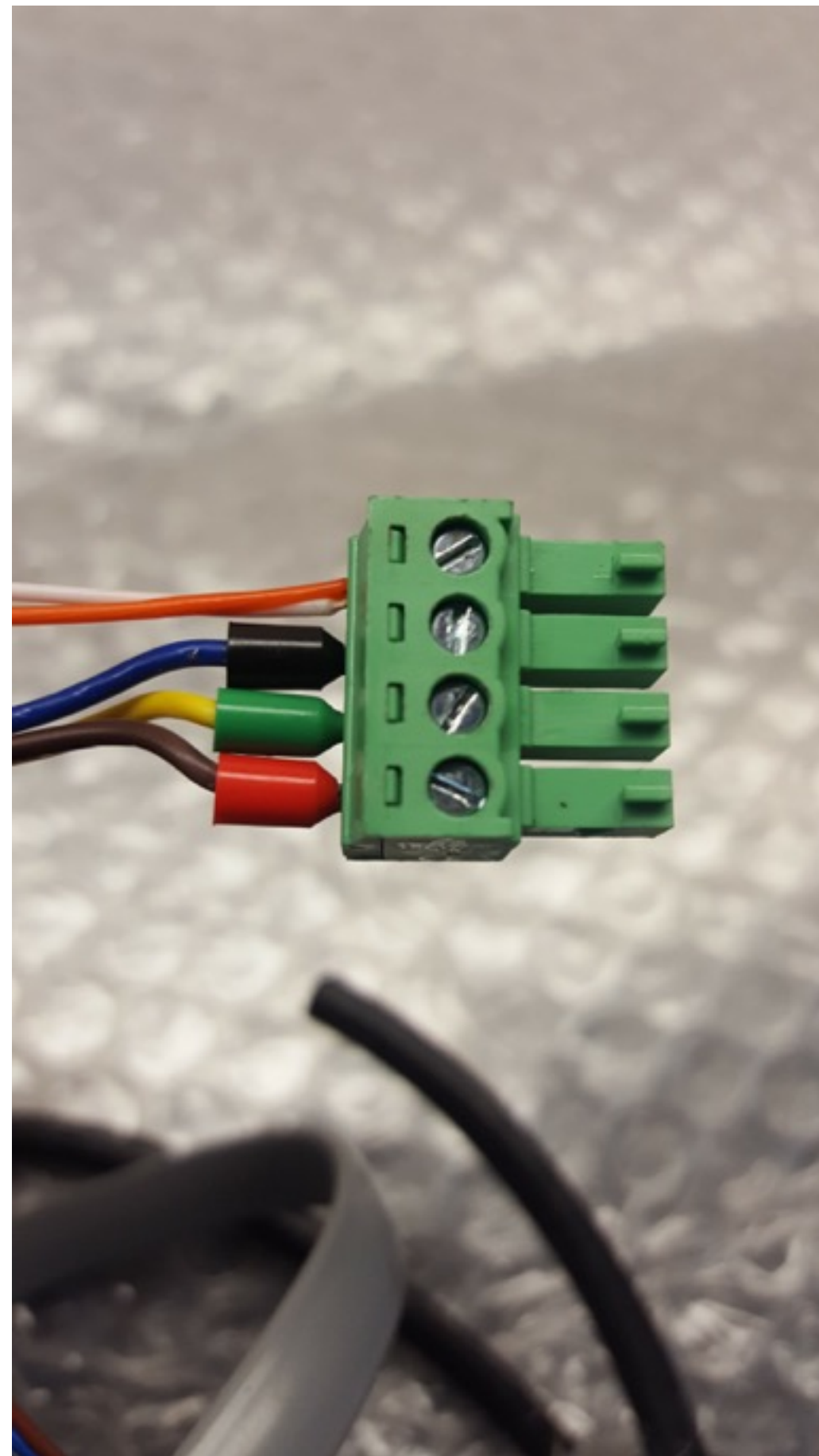
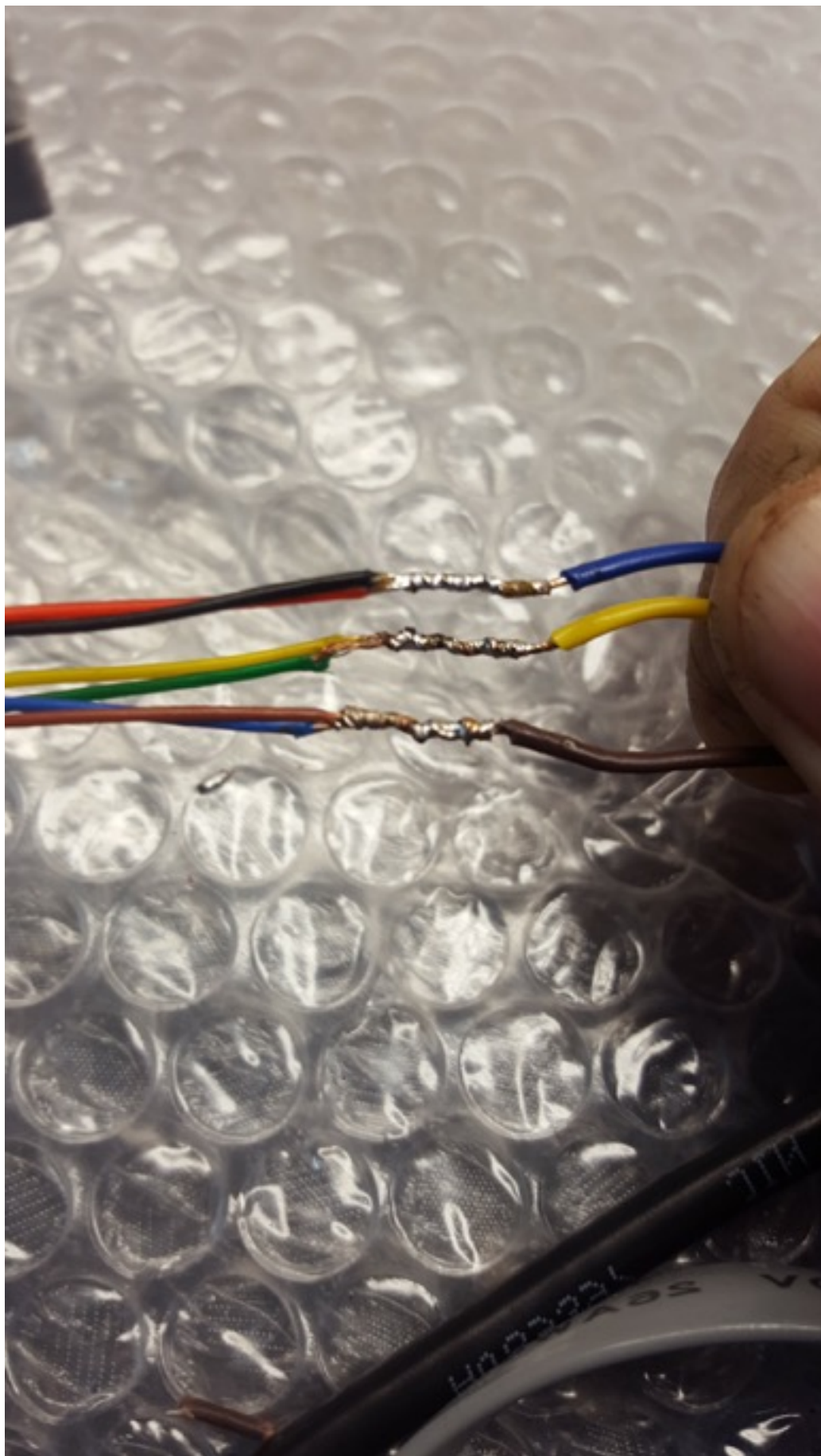
Pins	Define	Colour
1	Power supply output +5V	Orange
2	Power supply output +5V	White
3	RS-485-B	Black
4	RS-485-B	Red
5	RS-485-A	Green
6	RS-485-A	Yellow
7	Ground	Blue
8	Ground	Brown



The diagram illustrates the internal wiring of an 8-pin connector, likely an RJ-45 plug, showing the connection of four pairs of wires. The wires are labeled as follows:

- +5V**: Power supply output +5V (Orange)
- B**: RS-485-B (Black)
- A**: RS-485-A (Green)
- GND**: Ground (Blue)

The diagram shows the internal wiring of the connector, with the labels +5V, B, A, and GND indicating the function of each pair of wires. The wires are connected to the pins of the connector, which is shown in a cross-section view.



RS-485 Connection in 60 Amp Controller

